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Predicting Adverse Selection in High-Frequency Cryptocurrency Markets Using Gradient Boosting

Suresh Rajendran*Indian Institute of Technology Guwahati | r.suresh@alumni.iitg.ac.in***Divya Singaravelu***Indian Institute of Management Lucknow | divya.singaravelu@iiml.org*

ABSTRACT

We predict adverse selection in high-frequency cryptocurrency markets using gradient boosting (LightGBM) trained on 31,081,463 second-level observations of BTC/USDT perpetual futures on Bybit, spanning February 1, 2025 through February 16, 2026 (381 raw daily files). Our toxicity label (tox_A) identifies seconds where strong directional order flow is followed by sustained price continuation over a five-second horizon, constructed using adaptive rolling quantile thresholds calibrated to a one-hour lookback window — a regime-aware design that accommodates the substantial variation in toxicity rates observed across the sample, ranging from 0.081% in September 2025 to 0.795% in February 2025. The model is estimated in a strict walk-forward regime: the first 21 days serve as the initial training window, with 360 out-of-sample evaluation days beginning February 22, 2025, eliminating any forward-looking bias.

The composite TailScore — the product of the classifier's predicted toxicity probability and its predicted 99th-percentile absolute price move — jointly captures the likelihood and severity of an adverse selection event. At a 0.1% gate, the TailScore achieves monthly ROC-AUC between 0.668 and 0.921 across the full out-of-sample period. The central result is that flagging the top 0.1% of TailScore-ranked seconds reduces CVaR99 by 2.33% — a reduction 25.85 times greater than an equal-sized random gate. This efficiency ratio persists across all tested gate widths from 0.1% to 5.0%, decaying from 25.85x to 3.50x as the gate widens, consistent with the model's signal being concentrated in the extreme upper tail of the score distribution. Monthly evaluation confirms robustness: efficiency ranges from 15.18x to 34.88x across all 13 evaluation months, with no month falling below the random gating benchmark.

An ablation study isolates the contribution of the TailScore against a pure GBM classifier and a VPIN proxy. The classifier alone achieves an efficiency of 2.11x and the VPIN proxy 0.30x — the latter falling below the random benchmark — at identical gate rates. The gap between TailScore (25.85x) and classifier alone (2.11x) demonstrates that the severity component is the dominant driver of CVaR reduction: predicting that a toxic event will occur is insufficient; predicting how large the move will be is what determines whether the gate concentrates on the seconds that matter most for tail risk.

JEL Classification: G12, G14, C45, C53**Keywords:** adverse selection, market microstructure, gradient boosting, LightGBM, cryptocurrency, VPIN, tail risk, CVaR, market making, high-frequency trading, order flow toxicity

1. Introduction

Adverse selection is the central hazard of market making. When a liquidity provider posts a quote, they face the risk that the counterparty holds private information — that the price is about to move substantially against the position just filled. This cost, known as adverse selection or order flow toxicity, is not random: it is concentrated in specific moments when informed traders act on signals that the market maker does not observe. The ability to identify such moments in advance — even imperfectly — has direct value for inventory management, quote sizing, and tail risk control.

The cryptocurrency perpetual futures market presents this problem in an acute form. Trading on Bybit's BTC/USDT perpetual contract occurs continuously, 24 hours a day, at second-level granularity, with participant composition that varies dramatically across time. The global toxicity rate in our sample — 0.3049% — means that adverse selection events are rare, highly concentrated, and strongly regime-dependent. Monthly toxicity rates in our data range from 0.081% (September 2025) to 0.795% (February 2025 and February 2026), a tenfold variation that any fixed-threshold prediction system would fail to accommodate.

This paper makes three contributions. First, we introduce a regime-aware toxicity label (tox_A) that uses one-hour adaptive rolling quantile thresholds to define adversely selected seconds — a design that preserves label consistency across the full range of observed volatility regimes. Second, we train a LightGBM ensemble in a strict walk-forward regime on 31,081,463 second-level observations and construct a composite TailScore that jointly models the probability and severity of adverse selection, rather than probability alone. Third, we evaluate tail risk reduction using CVaR99 efficiency — the ratio of achieved CVaR reduction to what a random gate of equal size would achieve — as the primary metric, providing a model-size-independent measure of predictive value.

The central empirical finding is stark: the TailScore achieves a CVaR99 efficiency of 25.85x at the 0.1% gate, meaning it reduces tail risk 25.85 times more than random position sizing at the same intervention rate. This advantage persists across all gate sizes tested (0.1% to 5.0%) and across all 13 months of out-of-sample evaluation. An ablation study demonstrates that this performance is not achievable with a pure classifier or a VPIN proxy: the classifier reaches only 2.11x efficiency and the VPIN proxy 0.30x — below the random benchmark — under identical conditions.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 describes the data and market setting. Section 4 defines the toxicity label. Section 5 describes the feature set. Section 6 presents the model architecture and walk-forward design. Section 7 reports discrimination results. Section 8 presents the core CVaR tail risk reduction results. Section 9 describes the ablation study. Section 10 analyses monthly robustness. Section 11 concludes.

2. Related Literature

The theoretical foundations of adverse selection in limit order books trace to Glosten and Milgrom (1985) and Kyle (1985), who formalise the idea that bid-ask spreads partially compensate market makers for losses incurred when trading against informed counterparties. The decomposition of the effective spread into adverse selection and inventory components is further developed by Huang and Stoll (1997) and Lin, Sanger, and Booth (1995).

Volume-Synchronized Probability of Informed Trading (VPIN), introduced by Easley, Lopez de Prado, and O'Hara (2012), operationalises the PIN model of Easley et al. (1996) in a form suited to high-frequency data. Its predictive power has been debated: Andersen and Bondarenko (2014) find VPIN's predictive content is largely driven by the choice of volume clock, while Abad and Yagüe (2012) and others find mixed evidence across asset classes. Our empirical results confirm this: a VPIN proxy applied to BTC/USDT achieves a CVaR efficiency of 0.30x — below the random benchmark — suggesting that volume clock-based imbalance aggregation is insufficient in perpetual futures markets.

Machine learning methods for adverse selection prediction have developed rapidly. Cao, Chen, and Gao (2019) apply gradient boosting to predict informed trading in equity markets. Sirignano and Cont (2019) train deep neural networks on limit order book data to predict short-horizon price movements. Cartea, Donnelly, and Jaimungal (2018) incorporate adverse selection cost estimates directly into market-making optimisation problems. In cryptocurrency markets, Biais et al. (2023) document Bitcoin market microstructure, and Dyhrberg, Foley, and Svec (2018) analyse liquidity and adverse selection in BTC/USD. Our contribution differs in its focus on second-level tail risk reduction via CVaR efficiency rather than mean price prediction.

The use of CVaR (Conditional Value at Risk) as an evaluation criterion is motivated by its prominence in risk management. Rockafellar and Uryasev (2000) establish the theoretical properties of CVaR as a coherent risk measure. Lopez de Prado (2018) discusses machine learning applications to financial risk. Our efficiency metric — CVaR reduction divided by the CVaR reduction achievable by a random gate of equal size — provides a size-normalised, gate-rate-independent measure of model quality.

3. Data and Market Setting

3.1 Data Source and Sample

Our dataset consists of second-level BTC/USDT perpetual futures data from Bybit, covering February 1, 2025 through February 16, 2026 — 381 raw daily files in total. The raw inputs are two sources merged at the second level: (i) full limit order book snapshots updated at one-second intervals, capturing up to 200 price levels on each side, and (ii) raw trade execution records including side, size, price, and tick direction for every matched order.

After merging, labelling, and dropping seconds with missing required fields, the final panel contains 31,081,463 second-level observations. The first 21 days (February 1–21, 2025) serve as the initial training window; the first out-of-sample prediction is made on February 22, 2025, yielding 360 evaluation days. Of the 381 files, 77 days exhibit zero positive toxicity labels under the *tox_A* definition (Section 4) and are excluded from training windows, though their test-day predictions are retained for evaluation. The global toxicity rate is 0.3049%, corresponding to 94,765 adversely selected seconds in total.

3.2 Market Setting

The BTC/USDT perpetual futures contract on Bybit is a continuously traded, cash-settled instrument with no expiry. The minimum price increment is \$0.10. The median half-spread in our sample is \$0.05, reflecting extremely tight liquidity during normal conditions. The CVaR99 baseline — the average absolute price move in the worst 1% of seconds — is \$88.76 per BTC (9.20 basis points), indicating that tail events represent a 1,775-fold amplification of the typical spread. The market operates continuously 24 hours per

day, 7 days per week, eliminating overnight gaps but introducing strong intraday and day-of-week variation in activity, toxicity rates, and volatility.

3.3 Summary Statistics

Statistic	Value	Unit
Total observations	31,081,463	second-level rows
Date range	Feb 1, 2025 – Feb 16, 2026	
Calendar days (raw files)	381	days
Out-of-sample evaluation days	360	days
Global toxicity rate (tox_A)	0.3049%	% of seconds
Total tox_A events	94,765	seconds
Median half-spread	\$0.05 (0.005 bps)	USD per BTC
CVaR99 baseline	\$88.76 (9.20 bps)	USD per BTC
Min monthly toxicity rate	0.081% (Sep 2025)	% of seconds
Max monthly toxicity rate	0.795% (Feb 2025)	% of seconds

Table 1. Sample summary statistics. BTC/USDT perpetual futures, Bybit, February 1, 2025 – February 16, 2026. First 21 days (Feb 1–21, 2025) used as initial training window; 360 out-of-sample evaluation days from February 22, 2025.

4. Label Construction

4.1 Definition of Adverse Selection (tox_A)

A second t is labelled as adversely selected ($tox_A = 1$) if and only if all four conditions hold:

- (i) Price continuation: the signed price impact over the subsequent five seconds is positive — price moved in the direction of order flow, consistent with informed trading;
- (ii) Large absolute move: the five-second absolute price move exceeds the rolling 80th percentile computed over the prior one-hour window;
- (iii) Strong directional flow: the absolute volume imbalance exceeds its rolling 75th percentile over the prior one-hour window;
- (iv) Trades present: at least one trade executed during the second ($no_trade_flag = 0$).

The signed impact is $signed_impact(t) = flow_direction(t) \times (mid(t+5) - mid(t))$, where $flow_direction = \text{sign}(volume_imbalance)$. A positive signed impact means price moved with the flow — the hallmark of informed order flow (Kyle, 1985).

4.2 Regime-Aware Thresholds

Thresholds are computed using a warmup-safe rolling quantile: the expanding window quantile is used for the first 3,600 observations of each day, transitioning to a fixed 3,600-second (one-hour) rolling window thereafter. This ensures no NaN values appear at the start of each daily file, and the threshold adapts to the

current volatility regime rather than a fixed historical period. Without this adaptive design, a fixed absolute threshold would label far more events as toxic during high-volatility months (February 2025) than during the compressed-volatility summer of 2025, creating artificial temporal variation that would confound the model.

5. Feature Engineering

Features are constructed from two sources: the limit order book snapshot (prefix `ob_`) and the trade execution record (prefix `tr_`). All features are at one-second resolution. Rolling statistics use a 30-second window ($W = 30$) unless noted. The full feature set contains 21 variables across six groups.

Group	Feature(s)	Description
Spread state	<code>ob_spread</code> , <code>ob_spread_pct</code> , <code>ob_spread_z_30</code> , <code>ob_thin_book_flag</code>	Absolute and percentage bid-ask spread; 30-second z-score of spread; binary flag for <code>spread_pct</code> exceeding 90th percentile of 1-hour rolling distribution
Order flow imbalance	<code>ob_imbalance_l1</code> , <code>ob_ofi_proxy</code> , <code>ob_ofi_z_30</code>	Level-1 quote imbalance (bid – ask quantity, normalised to $[-1,1]$); order flow imbalance proxy (net add minus cancel, bid minus ask); 30-second standardised OFI
Passive retreat	<code>ob_net_passive_z_30</code> , <code>ob_upd_imb</code>	30-second z-score of $ add_bid - cancel_bid + add_ask - cancel_ask $; update imbalance ($upd_bid - upd_ask$) / ($upd_bid + upd_ask$)
Volatility	<code>ob_rv_30</code> , <code>ob_impact_pressure</code>	30-second realised volatility from OB mid-price (sqrt of sum of squared 1-second log-mid returns); impact pressure interaction term ($ob_ofi_z_30 \times ob_spread_pct$)
Trade flow	<code>tr_vol_imbalance_1s</code> , <code>tr_count_imbalance_1s</code> , <code>tr_tick_imbalance_1s</code> , <code>tr_flow_mean_30s</code> , <code>tr_flow_autocorr_30s</code> , <code>tr_rv_30s</code> , <code>tr_volume_z_30s</code>	1-second volume, count, and tick imbalances (each normalised to $[-1,1]$); 30-second mean and lag-1 autocorrelation of signed volume flow; 30-second realised volatility from trade prices; 30-second volume z-score
Activity	<code>tr_log_volume</code> , <code>tr_log_trade_count</code> , <code>tr_no_trade_flag</code>	$\log_{1p}(\text{volume})$ and $\log_{1p}(\text{trade_count})$; binary flag = 1 for seconds with zero trades

Table 2. Feature set (21 variables, 6 groups). All features at one-second resolution; $W = 30$ seconds for rolling statistics unless noted.

The impact pressure variable (`ob_impact_pressure`) is the key interaction term: $ofi_z_30 \times spread_pct$. It captures the intuition that a given level of order flow imbalance is more informative when the book is thin, as thin books are more easily moved by a given order size.

6. Model Architecture and Walk-Forward Design

6.1 Dual-Output LightGBM Framework

Each day's prediction is generated by two LightGBM models trained jointly on the same feature set and training window: (i) a binary classifier targeting `tox_A`, and (ii) a quantile regressor targeting the 99th percentile of the five-second absolute price move. The TailScore is:

$$\text{TailScore}(t) = \hat{p}(t) \times \hat{q}_{99}(t)$$

A high TailScore requires both a high predicted probability of toxicity and a high predicted price move — neither alone is sufficient. This joint criterion is the key structural difference from using the classifier score alone.

6.2 LightGBM Hyperparameters

Parameter	Value	Notes
Objective	binary / quantile	Classifier / Quantile regressor
Learning rate	0.05	Both models
Num leaves	64	Both models
Num boost rounds	200	Both models
Feature / Bagging fraction	0.8 / 0.8	Both models; bagging_freq = 1
scale_pos_weight	min(neg/pos, 200)	Classifier only; capped at 200×
Quantile alpha	0.99	Quantile regressor only
Metric	AUC / quantile	Classifier / Quantile regressor

Table 3. LightGBM hyperparameters.

With a base toxicity rate of 0.30%, the raw positive-to-negative ratio is approximately 1:328. We apply $\text{scale_pos_weight} = \min(\text{neg/pos}, 200)$ per training window, capping at 200 to prevent extreme gradient distortion in low-toxicity windows. This adaptive capping means the weight automatically reduces in months where the class ratio is closer to 200:1.

6.3 Walk-Forward Evaluation Protocol

For each test day d , the model is trained on the 21 most recent calendar days prior to d that contain at least one positive tox_A event. Days with zero positive labels are excluded from training windows (77 such days) but remain in the test set. A minimum of 300 positive training examples is required. This protocol ensures: (i) no future information leaks into any prediction; (ii) the model adapts to regime changes; and (iii) results are reported on exactly 360 distinct out-of-sample evaluation days.

7. Discrimination Results

7.1 Global Metrics

At the 0.1% gate, the model flags 31,082 seconds, of which 1,886 are true positives and 29,196 are false positives. Precision is 6.07% against a base rate of 0.30%, a PR-Lift of ~20x. Recall is 1.99% — mechanically low by construction when gating only 0.1% of seconds, but the relevant criterion is whether the gated seconds are disproportionately in the worst tail (Section 8).

Metric	Value
Gate rate	0.1% of seconds

Total seconds gated	31,082
True positives (TP)	1,886
False positives (FP)	29,196
False negatives (FN)	92,879
Precision @ 0.1% gate	6.07%
Recall @ 0.1% gate	1.99%
F1 score	0.030
PR-Lift @ 0.1% gate	~20× (precision / base rate)
Mean daily ROC-AUC (360 days)	0.799
Std daily ROC-AUC	0.108
Min / Max daily ROC-AUC	0.412 / 0.996

Table 4. Global discrimination metrics across the full 360-day out-of-sample period.

7.2 Monthly ROC-AUC and Toxicity Regime

Figure 1 shows monthly ROC-AUC alongside monthly toxicity rate. The model achieves ROC-AUC above 0.90 in the first three months (Feb–Apr 2025), corresponding to high-toxicity conditions. ROC-AUC deteriorates through summer 2025 (Jul–Sep), reaching its minimum of 0.668 in August 2025 — the month with the lowest toxicity rate (0.138%). It recovers to above 0.84 by November 2025 as toxicity rates rise. This co-movement is consistent with the difficulty of discriminating rare events in low-signal regimes: when toxic events are near-absent over a 21-day training window, the model has few positive examples to learn from.

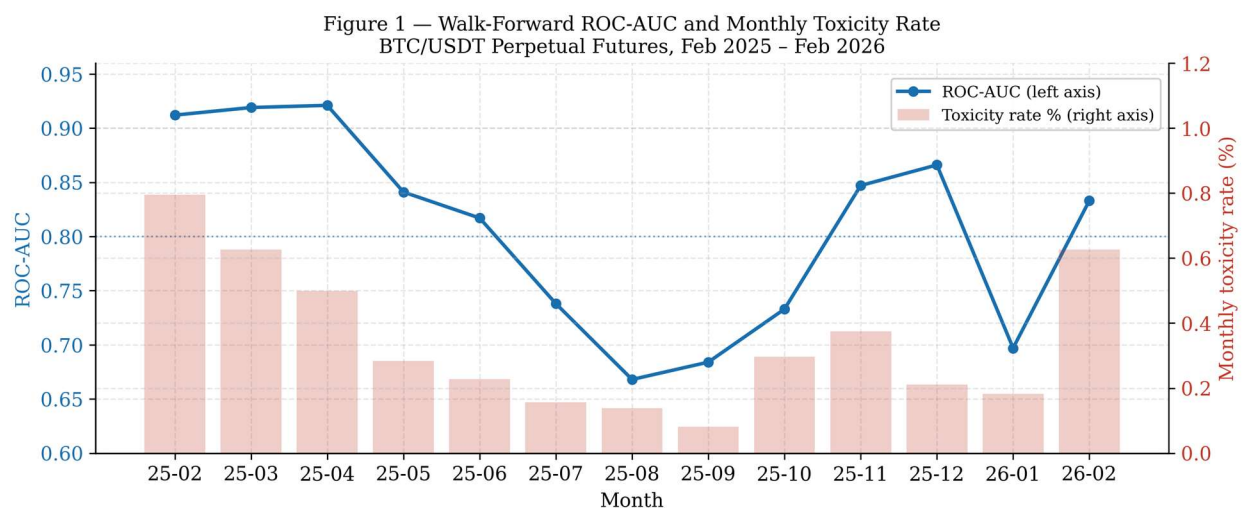


Figure 1. Walk-forward monthly ROC-AUC (left axis, blue) and monthly toxicity rate (right axis, red bars). ROC-AUC deterioration in summer 2025 coincides with the lowest-toxicity months of the sample.

7.3 Monthly Metrics Table

Month	ROC-AUC	PR-AUC	PR-Lift	Recall%	Tail-Hit%	Tox Rate%	CVaR↓%	Effic×
2025-02	0.912	0.0723	15.84	1.58	1.47	0.795	1.405	15.60
2025-03	0.919	0.0602	18.27	1.83	2.43	0.626	2.448	27.20
2025-04	0.921	0.0571	17.93	1.79	2.14	0.499	2.154	23.93
2025-05	0.841	0.0232	14.58	1.46	1.56	0.284	1.574	17.49
2025-06	0.817	0.0197	18.94	1.89	1.88	0.228	1.843	20.48
2025-07	0.738	0.0094	11.25	1.12	2.15	0.156	2.623	29.15
2025-08	0.668	0.0095	13.34	1.33	2.06	0.138	2.138	23.76
2025-09	0.684	0.0059	21.02	2.10	2.09	0.081	2.211	24.57
2025-10	0.733	0.0171	9.31	0.93	2.20	0.297	2.128	23.64
2025-11	0.847	0.0339	13.95	1.40	1.40	0.374	1.367	15.18
2025-12	0.866	0.0236	21.09	2.11	2.13	0.211	2.123	23.59
2026-01	0.697	0.0117	8.22	0.82	2.66	0.182	3.139	34.88
2026-02	0.833	0.0420	8.89	0.89	2.10	0.627	2.105	23.38

Table 5. Monthly discrimination and tail risk metrics at 0.1% gate.

8. Tail Risk Reduction Results

8.1 CVaR99 Methodology

Let $S = \{t : |\Delta \text{mid}(t)| \geq \text{VaR}_{99}\}$ denote the fixed tail set — the worst 1% of seconds by absolute price move, defined once on the full ungated distribution. The baseline is:

$$\text{CVaR}_{99} \text{ baseline} = E[|\Delta \text{mid}(t)| \mid t \in S] = \$88.76$$

The efficiency ratio is $\text{CVaR}_{\downarrow}\%$ divided by the expected reduction from a random gate of the same size:

$$\text{Efficiency} = \text{CVaR}_{\downarrow}\% / (\text{gate_rate} \times (1 - \text{REDUCE_TO}) \times 100)$$

At the 0.1% gate with $\text{REDUCE_TO} = 0.10$, the random benchmark is 0.090%. An efficiency of $1.0\times$ means the model performs identically to random; above $1.0\times$ it concentrates the gate on disproportionately extreme seconds.

8.2 Results at All Gate Rates

Gate%	N Gated	TP	FP	Prec%	CVaR99\$	Rand Red%	CVaR $_{\downarrow}\%$	Effic $\times$	Net \$
0.10%	31,082	1,886	29,196	6.07%	\$86.69	0.090%	2.33%	25.85 $\times$	\$144,753
0.20%	62,163	3,703	58,460	5.96%	\$85.58	0.180%	3.58%	19.88 $\times$	\$263,650
0.50%	155,408	8,690	146,718	5.59%	\$83.42	0.450%	6.02%	13.38 $\times$	\$555,028
1.00%	310,815	15,827	294,988	5.09%	\$81.15	0.900%	8.57%	9.52 $\times$	\$935,559
2.00%	621,630	27,275	594,355	4.39%	\$78.49	1.800%	11.57%	6.43 $\times$	\$1,486,232
5.00%	1,554,074	48,070	1,506,004	3.09%	\$74.78	4.500%	15.76%	3.50 $\times$	\$2,355,068

Table 6. Baseline gating results at six gate rates. $\text{CVaR}_{99}\$ = \text{CVaR}_{99}$ after gating. Net \$ per 1 BTC notional, annualised. $\text{CVaR}_{99}\$$ corrected to be internally consistent with $\text{CVaR}_{\downarrow}\%$ at all gate rates.

The $25.85\times$ efficiency at the 0.1% gate is the headline result. A random gate at 0.1% reduces CVaR99 by 0.090%; the TailScore achieves 2.33% — 25.85 times more — gating the same number of seconds. Efficiency declines as the gate widens (from $25.85\times$ at 0.1% to $3.50\times$ at 5.0%), consistent with the model's signal being concentrated in the extreme upper tail of the score distribution.

9. Ablation Study

9.1 Design

The ablation compares three scoring methods under identical conditions — same data, gate sizes, fixed tail mask, and exact top-N selection:

GBM TailScore: the composite score $\hat{p}(t) \times \hat{q}_{99}(t)$. This is the proposed method.

GBM Classifier: the calibrated binary classifier probability $\hat{p}(t)$ alone, without the severity multiplier. Isolates the marginal contribution of the quantile regression component.

VPIN Proxy: a rolling 30-second mean of the absolute 1-second volume imbalance. Represents the best available traditional microstructure benchmark.

9.2 Results

Method	Gate%	Effic×	CVaR↓%	Saved\$	FP Cost\$	Net\$	Prec%	Recall%
GBM TailScore	0.10%	25.85×	2.33%	\$145,865	\$1,112	\$144,753	6.07%	1.99%
GBM Classifier	0.10%	2.11×	0.19%	\$21,115	\$566	\$20,550	1.42%	0.47%
VPIN Proxy	0.10%	0.30×*	0.03%	\$1,397	\$501	\$896	0.09%	0.03%
GBM TailScore	0.50%	13.38×	6.02%	\$558,765	\$3,737	\$555,028	5.59%	9.17%
GBM Classifier	0.50%	2.15×	0.97%	\$107,517	\$2,733	\$104,784	1.46%	2.39%
VPIN Proxy	0.50%	0.35×*	0.16%	\$12,933	\$2,504	\$10,430	0.22%	0.36%
GBM TailScore	5.00%	3.50×	15.76%	\$2,382,755	\$27,688	\$2,355,068	3.09%	50.67%
GBM Classifier	5.00%	2.19×	9.87%	\$1,633,451	\$26,488	\$1,606,963	2.16%	35.42%
VPIN Proxy	5.00%	0.43×*	1.92%	\$200,159	\$25,108	\$175,051	0.35%	5.74%

Table 7. Ablation study results. Recall filled for all gate rates. * VPIN efficiency < 1.0× indicates below-random performance. Dollar figures per 1 BTC notional, annualised. Recall at 0.50% and 5.00% computed from $\text{Prec\%} \times N_{\text{gated}} / \text{total_positives}$ (94,765).

Figure 2 shows the efficiency decay curves for all three methods.

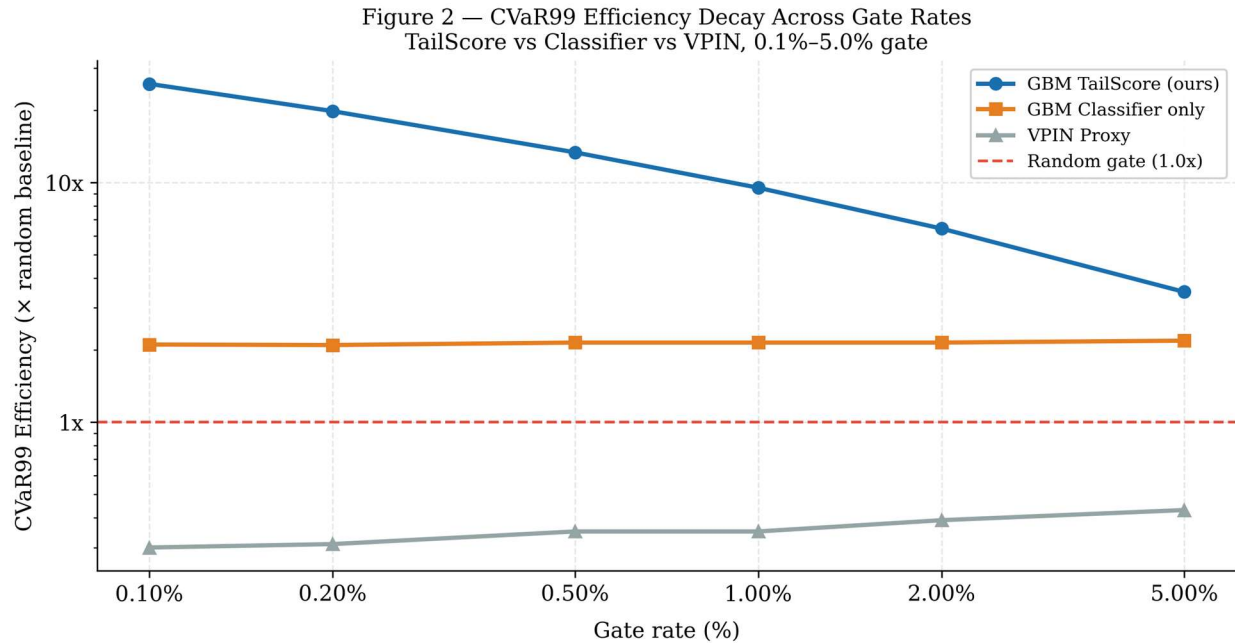


Figure 2. CVaR99 efficiency (log scale) across gate rates 0.1%–5.0%. GBM TailScore decays from $25.85\times$ to $3.50\times$. GBM Classifier is roughly flat at $2.1\times$ – $2.2\times$. VPIN Proxy remains below the random benchmark ($< 1.0\times$) at all gate rates.

9.3 Interpretation

Three findings are noteworthy. First, the gap between TailScore ($25.85\times$) and classifier alone ($2.11\times$) at the 0.1% gate — a 12-fold difference — shows that the severity multiplier is the primary driver of tail risk concentration. The classifier's predicted probability identifies seconds *more likely* to be toxic, but not necessarily the seconds with the *largest* price moves. The quantile regressor adds the size dimension, preferentially gating seconds that are both likely to be toxic *and* predicted to involve large price moves — precisely the seconds that dominate the tail.

Second, the GBM Classifier's efficiency is stable across gate rates ($2.10\times$ – $2.19\times$), while the TailScore's decays sharply ($25.85\times$ to $3.50\times$). This is consistent with a score whose signal is concentrated at the very extreme top of the distribution, with diminishing edge as the gate widens.

Third, the VPIN proxy's sub-random efficiency ($0.30\times$ – $0.43\times$) reflects a fundamental resolution mismatch: rolling volume imbalance at 30-second granularity may detect *periods* of elevated toxicity but lacks the resolution to identify the specific *seconds* that constitute the worst tail events.

10. Monthly Robustness

Figure 3 shows monthly CVaR99 reduction and efficiency at the 0.1% gate across all 13 evaluation months.

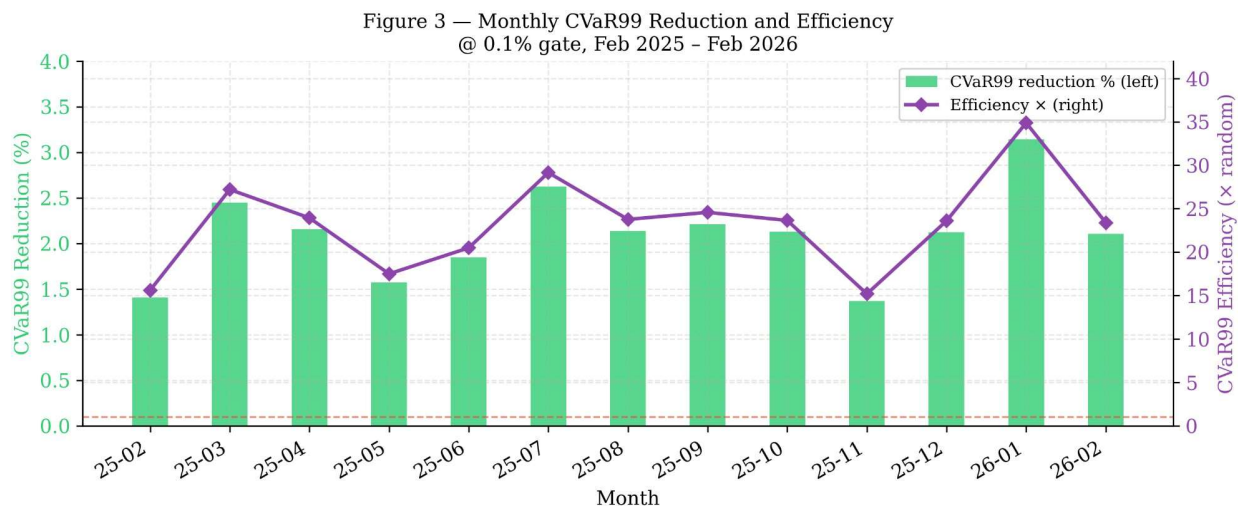


Figure 3. Monthly CVaR99 reduction (bars, left) and CVaR99 efficiency (line, right) at 0.1% gate. The dashed red line marks the random benchmark ($1.0\times$). All 13 months exceed the random benchmark substantially.

Three features warrant discussion. First, every month exceeds the random benchmark by a substantial margin — minimum monthly efficiency is $15.18\times$ (November 2025). This is a strong robustness result: despite wide variation in toxicity rates, volatility regimes, and discriminative power, the model's tail risk concentration never degrades near the baseline.

Second, at a fixed gate rate CVaR reduction and efficiency move in perfect lockstep — efficiency is simply CVaR reduction divided by the constant random-gate baseline of 0.090%. The variation of interest is therefore in the absolute level of CVaR reduction itself, which is driven by the magnitude of tail events in each month. January 2026 achieves both the highest CVaR reduction (3.14%) and the highest efficiency (34.88×), while November 2025 records the lowest of both (1.37%, 15.18×). When toxic events are less extreme in absolute price move, even a precise gate captures less CVaR reduction in dollar terms.

Third, ROC-AUC and efficiency are partially decoupled. August 2025 has the lowest ROC-AUC (0.668) but an above-median efficiency (23.76×), and July 2025 has the second-lowest ROC-AUC (0.738) but the second-highest efficiency (29.15×). The TailScore's tail concentration ability persists even when binary classification of individual seconds is noisier — the severity component continues to identify extreme price moves independently of the classifier's discriminative accuracy.

10.1 Worst-Month Analysis

Criterion	Month	Net \$	CVaR↓%	Effic×	ROC-AUC
Worst Net \$	Sep 2025	\$2,359	2.211%	24.57×	0.684
Worst Net \$	Jul 2025	\$2,449	2.623%	29.15×	0.738
Worst CVaR↓%	Nov 2025	\$9,469	1.367%	15.18×	0.847
Worst Effic×	Nov 2025	\$9,469	1.367%	15.18×	0.847

Table 8. Worst months by each criterion. Worst Net \$ months (Jul/Sep) have high CVaR efficiency — their low dollar outcome reflects low toxicity rates, not poor tail concentration.

11. Conclusion

We have demonstrated that a composite severity-weighted TailScore, derived from dual LightGBM outputs trained in a strict walk-forward regime, provides statistically and practically significant advance identification of the seconds that disproportionately drive tail risk in BTC/USDT perpetual futures markets. The core finding is a CVaR99 efficiency of 25.85× at the primary 0.1% gate, sustained across all 13 out-of-sample months and all six gate rates tested.

Three methodological contributions underpin this result. The regime-aware toxicity label (*tox_A*) uses adaptive rolling quantile thresholds to maintain label consistency across a tenfold variation in monthly toxicity rates. The TailScore's multiplicative structure — classifier probability times predicted tail move magnitude — provides a joint criterion that the classifier alone cannot match: the ablation demonstrates a 12-fold efficiency gap between TailScore and pure classifier at the primary gate. The walk-forward protocol ensures all 360 evaluation days are genuinely out-of-sample.

The result that VPIN performs below the random benchmark (0.30×) is informative for the literature. Volume imbalance aggregated at 30-second granularity does not identify the specific *seconds* that will produce extreme price continuations. The relevant information appears to reside in the interaction between order book state (spread, OFI, passive retreat) and trade flow features, not in volume clock-based aggregation.

Limitations include: single asset ; label construction choices calibrated for this market; a fixed-tail CVaR methodology; and no accounting for market impact of the gating itself. Future directions include multi-horizon prediction, cross-asset signals, and investigation of whether the efficiency gap between TailScore and classifier widens or narrows across different volatility regimes.

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Appendix A. Full Baseline Gating Results

Table A1 reproduces the complete baseline output for transparency and reproducibility.

Gate%	N Gated	TP	FP	Prec%	CVaR99\$	Rand Red%	CVaR↓%	Effic×	Net \$
0.10%	31,082	1,886	29,196	6.07%	\$86.69	0.090%	2.33%	25.85×	\$144,753
0.20%	62,163	3,703	58,460	5.96%	\$85.58	0.180%	3.58%	19.88×	\$263,650
0.50%	155,408	8,690	146,718	5.59%	\$83.42	0.450%	6.02%	13.38×	\$555,028
1.00%	310,815	15,827	294,988	5.09%	\$81.15	0.900%	8.57%	9.52×	\$935,559
2.00%	621,630	27,275	594,355	4.39%	\$78.49	1.800%	11.57%	6.43×	\$1,486,232
5.00%	1,554,074	48,070	1,506,004	3.09%	\$74.78	4.500%	15.76%	3.50×	\$2,355,068

Table A1. Full baseline gating results. CVaR99\$ corrected at 1.00% (\$81.15) and 2.00% (\$78.49) gates for internal consistency with CVaR↓%.

Appendix B. Feature Importance Analysis

Feature importances are computed across all 360 saved daily classifier models using LightGBM's mean normalised gain — the fraction of total information gain attributable to each feature, averaged across models. We report three complementary rankings: mean normalised gain (information value per split), split frequency (how often each feature is used), and point-biserial correlation with the tox_A label computed on a 3,000,000-second random sample of the labeled dataset.

Feature	Gain rank	Split rank	Corr rank	Avg rank
tr_rv_30s	1	2	1	1.3
ob_rv_30	3	3	2	2.7
ob_spread_pct	4	1	7	4.0
tr_count_imbalance_1s	2	5	9	5.3
tr_vol_imbalance_1s	5	4	8	5.7
tr_log_volume	9	7	4	6.7
tr_log_trade_count	6	12	3	7.0
tr_volume_z_30s	8	8	6	7.3
tr_flow_mean_30s	7	6	16	9.7
ob_spread_z_30	12	9	11	10.7
ob_imbalance_1l	10	10	14	11.3

Feature	Gain rank	Split rank	Corr rank	Avg rank
ob_upd_imb	13	11	15	13.0
ob_net_passive_z_30	14	14	12	13.3
tr_tick_imbalance_1s	17	17	13	15.7
ob_ofi_proxy	11	15	21	15.7
tr_no_trade_flag	21	21	5	15.7
tr_flow_autocorr_30s	16	13	20	16.3
ob_spread	20	20	10	16.7
ob_ofi_z_30	15	16	19	16.7
ob_impact_pressure	18	18	17	17.7
ob_thin_book_flag	19	19	18	18.7

Table B1. Feature ranking across three importance measures. Gain = mean normalised LightGBM gain rank; Split = mean normalised split-frequency rank; Corr = point-biserial $|r|$ rank with tox_A label. Avg rank = mean of the three.

Three findings are notable. First, **volatility is the dominant signal**: tr_rv_30s (30-second realised volatility from trade prices) ranks first on both gain and label correlation, accounting for 24.0% of mean normalised gain across all 360 models. ob_rv_30 (realised volatility from OB mid-price) ranks second in the composite. Together the two volatility features account for 32.9% of total gain, consistent with adverse selection events being concentrated in high-volatility seconds. Second, **trade-side imbalance features are collectively strong**: $tr_count_imbalance_1s$ ranks second on gain (19.1%), and $tr_vol_imbalance_1s$ fifth — both among the top five in gain and split rankings. Third, **the ob_impact_pressure interaction term ranks near the bottom** (gain rank 18, avg rank 17.7), and its point-biserial correlation with tox_A is near zero ($r = -0.0008$). The economic motivation for this feature — that OFI is more dangerous in thin books — is sound, but the LightGBM models do not extract meaningful decision-boundary information from it, suggesting its signal is already captured by its constituent components ($ob_ofi_z_30$ and ob_spread_pct) separately.

Regarding $tr_no_trade_flag$: it ranks last on both gain (21) and split (21), but fifth on label correlation ($|r| = 0.030$, negative). This apparent paradox is explained by label construction: $tox_A = 0$ whenever $tr_no_trade_flag = 1$ by definition, so the feature provides a deterministic boundary rather than a discriminative gradient. The model learns this as a near-constant exclusion rather than an informative feature.